AI Studio Challenge Project & Dataset Guidance

AI Studio Challenge Projects are real-world machine learning projects presented by industry partner organizations. A central part of the Break Through Tech AI Program, they are designed to give our student participants ("fellows") an experience comparable to an undergraduate-level technical internship project.

The following guidance will help you come up with a suitable and meaningful AI Studio Challenge Project on behalf of your company or organization. Once accepted, your project – along with a volunteer Challenge Advisor from your organization – will be assigned to a team of fellows to work on from August to December 2026.

Project Essentials

Your proposed project should:

- Be technical in nature and involve the use of **Machine Learning** techniques to address a real-world business or social challenge
- Ideally result in project artifacts or outcomes that your organization can actually make use of (e.g., an initial ML model for an engineering team initiative)
- Be solvable primarily using **Python** (this is the one programming language all Break Through fellows have shared knowledge of)
- Build on what our fellows learned in their ML Foundations summer course, while also presenting new and feasible technical challenges
- Allow the student team to engage in multiple stages of the ML project lifecycle, such as: business understanding, data preparation, modeling, and evaluation (deployment is not required)
- Be appropriate for a team of CS majors or other technical undergraduate students working in a part-time, unpaid, extracurricular capacity (~3 hours per week over ~3 months)
- Be **Open Source** in nature (e.g., not confidential for your organization; suitable for fellows to discuss publicly on GitHub and LinkedIn)

Problem Definition

<u>Upon submission, your proposed project must include</u>:

- A specific and clearly-articulated prompt, including the project's goal, business relevance, and success criteria (e.g., "In this project, you will use [type of data] and [ML techniques] to [target or outcome; e.g., build a model to predict something]. This will help our company address [business problem].")
- A dataset(s) that is either publicly available or provided by your organization *(see below for more details on Dataset requirements)*
- Specific monthly milestones that break down the project work into distinct phases (e.g., September: Data exploration and preprocessing)

Dataset

Your Challenge Project must include a robust and usable dataset(s) that the student team can access and use as soon as AI Studio kicks off in mid-August. The team should not be asked to find their own data or scrape data from public sources. You may, however, point them to additional data sources throughout the program.

The data you provide must meet the following criteria, and should be reviewed and vetted by you in advance to ensure quality and usability:

- Dataset must be publicly available (e.g., from Kaggle, Hugging Face, Data.gov, or other reputable public repositories), or provided by your organization
- Dataset must not include any confidential, proprietary, or sensitive information. All Personal Identifiable Information (PII) must be removed, anonymized, or redacted prior to sharing. Any other information not suitable for student team or public visibility must be removed or appropriately redacted or anonymized.
- Dataset must be accessible to the student team without special credentials or permissions. The student team will not be permitted to sign an NDA.
- Dataset must contain documentation (e.g., a data dictionary with feature definitions and data types; information on any preprocessing already performed; known limitations or quirks)

- Dataset must have a sufficient sample size, reasonable representation across categories or groups, and enough structure and complexity to support meaningful analysis
- Dataset should not be greater than 10 GB (to avoid resource or processing challenges)
- Dataset should be optimized for efficient storage and use in no-cost or low-resource environments
- Dataset should be in a machine-readable format (e.g., CSV, JSON)
- Dataset should ideally not be synthetically generated with AI or by other means, to avoid quality issues
- Dataset should require a reasonable amount of preprocessing and cleaning by the student team

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Resource Considerations

Your Challenge Project should be feasible using the free tier of Google Colab (or an equivalent no-cost environment where GPUs and storage can be accessed). The student team will not necessarily have independent access to compute or storage resources, and Break Through Tech is not able to provide such resources.

- If your project requires additional compute beyond what's available through the free tier of Google Colab or similar, your organization must be able to provide access at no cost to the student team (i.e., GPU or TPU credits)
- Projects that require significant resources and/or a deployment component must be paired with organization-provided resources (e.g., AWS, API keys, platform hosting), at no cost to the student team
- LLM-related projects must include organization-provided API keys.